

Institute of Actuaries of India

Subject CT6 – Statistical Methods

May 2014 Examinations

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1 :**(i) Lundberg's equality:**

Lundberg's inequality states that:

$$\psi(U) \leq \exp\{-RU\}$$

where U is the insurer's initial surplus and $\psi(U)$ is the probability of ultimate ruin.

R is a parameter associated with a surplus process known as the adjustment coefficient. Its value depends upon the distribution of aggregate claims and on the rate of premium income.

R can be interpreted as an (inverse) measuring risk. The larger the value of R , the smaller the upper bound for $\psi(U)$ will be. Hence, $\psi(U)$ would be expected to decrease as R increases.

R is a function of the parameters that affect the probability of ruin.

[4]

(ii) The surplus is:

$U(1) = 0.5 + 1 - S(1) = 1.5 - S(1)$, $U(t)$ = net road clearance fund (in thousands of Rs) & $S(t)$ denotes the aggregate cost incurred till end of the day.

$$U(2) = 0.5 + 1 * 2 - S(2) = 2.5 - S(2)$$

Considering the probability of non-ruin we require:

$$S(1) < 1.5 \text{ and } S(2) < 2.5$$

The probability of above is calculated in table below:

1 st period		2 nd period		Probability
Number of claims	Amount of claims	Number of claims	Amount of claims	
0 claims	0	0	0	$(e^{-0.5})(e^{-0.5}) = 0.36788$
		1 claim	1	$(e^{-0.5})(0.5e^{-0.5} * \frac{1}{2}) = 0.09197$
		1 claim	2	$(e^{-0.5})(0.5e^{-0.5} * \frac{1}{4}) = 0.04598$
		2 claims	1,1	$(e^{-0.5})(0.5^2/2 * e^{-0.5} * 0.5^2) = 0.01150$
1 claim	1	0 claims	0	$(0.5e^{-0.5} * \frac{1}{2})(e^{-0.5}) = 0.09197$
		1 claim	1	$(0.5e^{-0.5} * \frac{1}{2})(0.5e^{-0.5} * \frac{1}{2}) = 0.02299$

$$\text{Hence } P[U(t) > 0 \text{ for } t = 1 \text{ or } 2] = 0.6323 \Rightarrow P[U(t) < 0 \text{ for } t = 1 \text{ or } 2] = 0.3677$$

[8]

[Total Marks-12]

Solution 2 :

$$\begin{aligned}
 \text{(i)} \quad f(x) &= \frac{\alpha \lambda^\alpha}{(\lambda+x)^{(\alpha+1)}} \\
 \int_R^\infty x f(x) dx &= \int_R^\infty x \frac{\alpha \lambda^\alpha}{(\lambda+x)^{(\alpha+1)}} dx \\
 \text{substitute } t &= \lambda + x \\
 &= \int_{\lambda+R}^\infty (t-\lambda) \alpha \lambda^\alpha t^{-(\alpha+1)} dt \\
 &= \int_{\lambda+R}^\infty t^{-\alpha} \alpha \lambda^\alpha dt - \int_{\lambda+R}^\infty \lambda \alpha \lambda^\alpha t^{-(\alpha+1)} dt \\
 &= \alpha \lambda^\alpha \left[\frac{t^{-\alpha+1}}{-\alpha+1} \right]_{\lambda+R}^\infty - \alpha \lambda^{\alpha+1} \left[\frac{t^{-\alpha}}{-\alpha} \right]_{\lambda+R}^\infty \\
 &= \alpha \lambda^\alpha \left[-\frac{(\lambda+R)^{-\alpha+1}}{-\alpha+1} \right] - \alpha \lambda^{\alpha+1} \left[-\frac{(\lambda+R)^{-\alpha}}{-\alpha} \right] \\
 &= \frac{\alpha \lambda^\alpha}{(\alpha-1)(\lambda+R)^{\alpha-1}} - \frac{\lambda^{\alpha+1}}{(\lambda+R)^\alpha} \\
 &= \frac{\alpha \lambda^\alpha (\lambda+R) - \lambda^{\alpha+1} (\alpha-1)}{(\alpha-1)(\lambda+R)^\alpha} \\
 &= \frac{\alpha R \lambda^\alpha + \lambda^{\alpha+1}}{(\alpha-1)(\lambda+R)^\alpha} \\
 &= \frac{\lambda^\alpha (\alpha R + \lambda)}{(\alpha-1)(\lambda+R)^\alpha}
 \end{aligned}$$

[5]

(ii)

(a) Finding values of α and λ .

$$E(X) = \frac{\lambda}{\alpha-1} = 4.5$$

$$V(X) = \frac{\alpha \lambda^2}{(\alpha-2)(\alpha-1)^2} = 60.75$$

$$\Rightarrow \frac{\alpha}{(\alpha-2)} * (E(X))^2 = 60.75$$

$$\Rightarrow \frac{\alpha}{(\alpha-2)} = \frac{60.75}{20.25} = 3$$

$$\Rightarrow \alpha = 3$$

$$\lambda = 4.5 (\alpha-1) = 4.5*(3-1), \text{ thus } \lambda = 9$$

$$P[X > 25] = \left(\frac{\lambda}{\lambda+x} \right)^\alpha = \left(\frac{9}{9+25} \right)^3 = 0.018548$$

$$P[X > 30] = \left(\frac{\lambda}{\lambda+x} \right)^\alpha = \left(\frac{9}{9+30} \right)^3 = 0.012289$$

(b) Expected amount cede to reinsurer = $\int_R^{\infty} xf(x)dx - RP(X > R)$

$$= \frac{\lambda^{\alpha}(\alpha R + \lambda)}{(\alpha-1)(\lambda+R)^{\alpha}} - RP(X > R)$$

Retention = 25 $\Rightarrow = \frac{9^3(3+25+9)}{(2)(9+25)^3} - 25P(X > 25) = 0.779005 - 25 * 0.01855$

$$= 0.315225, \text{ i.e. INR 315.2 ceded to reinsurer}$$

Retention = 30 $\Rightarrow = \frac{9^3(3+30+9)}{(2)(9+30)^3} - 30P(X > 30) = 0.60833 - 30 * 0.012289$

$$= 0.23966, \text{ i.e. INR 239.7 ceded to reinsurer}$$

Expected number of claims = $500 * 15\% = 75$ claims

Reinsurance Policy 1 cedes $\text{INR } 315.2 * 75 = \text{INR } 23640$ to reinsurer

Reinsurance Policy 2 cedes $\text{INR } 239.7 * 75 = \text{INR } 17977.5$ to reinsurer

Insurer's cost would be the Premium less the claims ceded by the insurer.

Reinsurance Policy 1 would cost: $22000 - 23640 = (1640)$

Reinsurance Policy 2 would cost: $16000 - 17977.5 = (1977.5) \Rightarrow$ Profit more than policy 1

Thus Policy 2 is a better value for money.

[10]

[Total Marks-15]

Solution 3 :

(i) $M_S(t) = E(e^{tS})$

$$= E(E(e^{t(X_1 + X_2 + \dots + X_N)} | N))$$

$$= E(M_X(t)^N); \text{ Since the } X_i \text{ are independent and identically distributed}$$

$$= E(e^{N \log M_X(t)})$$

$$= M_N(\log M_X(t))$$

$$= \exp(\lambda \exp(\log(M_X(t)) - 1))$$

$$= \exp(\lambda (M_X(t) - 1))$$

[4]

(ii) $M'_S(t) = M_S(t) * \lambda M'_X(t)$

$$E(S) = M'_S(0) = M_S(0) * \lambda * M'_X(0)$$

$$= 1 * \lambda * \mu$$

$$= \lambda \mu$$

$$M''_S(t) = M'_S(t) * \lambda M'_X(t) + M_S(t) * \lambda M''_X(t)$$

$$E(S^2) = M''_S(0) = M'_S(0) * \lambda M'_X(0) + M_S(0) * \lambda M''_X(0)$$

$$= \lambda \mu * \lambda * \mu + 1 * \lambda * (\sigma^2 + \mu^2)$$

$$= \lambda^2 \mu^2 + \lambda \mu^2 + \lambda \sigma^2$$

$$\Rightarrow V(S) = E(S^2) - E(S)^2 = \lambda^2 \mu^2 + \lambda \mu^2 + \lambda \sigma^2 - \lambda^2 \mu^2$$

$$= \lambda * (\mu^2 + \sigma^2)$$

[6]

(iii) First, we must calculate the mean and variance of a single claim, Say Y.

Let us denote by X the underlying loss. Then

$$\begin{aligned} E(Y) &= \int_0^{200} 0.01 x e^{-1/100x} dx + \int_{200}^{\infty} (x + 50) * 0.01 * e^{-0.01x} dx \\ &= \int_0^{\infty} 0.01 x e^{-0.01x} dx + 50 \int_{200}^{\infty} 0.01 e^{-0.01x} dx \\ &= E(X) + 50 * P(X > 200) \\ &= 100 + 50 * e^{-\frac{200}{100}} \\ &= 100 + 6.7667 \\ &= 106.7667 \end{aligned}$$

$$\begin{aligned} E(Y^2) &= \int_0^{200} 0.01 x^2 e^{-0.01x} dx + \int_{200}^{\infty} (x + 50)^2 * 0.01 * e^{-0.01x} dx \\ &= \int_0^{\infty} 0.01 x^2 e^{-0.01x} dx + \int_{200}^{\infty} x e^{-0.01x} dx + 50^2 \int_{200}^{\infty} 0.01 e^{-0.01x} dx \\ &= E(X^2) + [-100xe^{-0.01x}] + \int_{200}^{\infty} 100 e^{-0.01x} dx + 50^2 P(X > 200) \\ &= 100^2 + 100^2 + 20000 e^{-2} + [-100^2 e^{-0.01x}] + 2500e^{-2} [2] \\ &= 20000 + 20000 e^{-2} + 10000 e^{-2} + 2500e^{-2} \\ &= 20000 + 32500 e^{-2} \\ &= 24398.3967 \end{aligned}$$

Thus, $E(S) = \lambda E(X) = 500 * 106.7667 = 53,383.38$

And, $V(S) = \lambda E(X^2) = 500 * 24398.3967 = 12,199,198.36$

[10]

[Total Marks-20]

Solution 4 :

(i)

Let D be the deductible, which means $\int_0^D f(x) dx = 0.2$

$$0.20 = \int_0^D \frac{\alpha \lambda^\alpha}{(\lambda+x)^{\alpha+1}} dx = \left[-\frac{\lambda^\alpha}{(\lambda+x)^\alpha} \right] = 1 - \frac{\lambda^\alpha}{(\lambda+D)^\alpha}; \text{ substituting values}$$

$$0.20 = 1 - \frac{300^\alpha}{(300+D)^\alpha}$$

$$\Rightarrow \frac{(300+D)}{300} = 0.80^{-1/3} \Rightarrow D = 300 (0.80^{-1/3} - 1) = 23.165$$

[4]

(ii)

The average net claim is given by $E[X - D | X > D]$

$$\begin{aligned} \int_D^\infty (x - D)f(x)dx &= \int_D^\infty \frac{x\alpha\lambda^\alpha}{(\lambda+x)^{\alpha+1}} dx - D \int_D^\infty \frac{\alpha\lambda^\alpha}{(\lambda+x)^{\alpha+1}} dx \\ &= \left[-\frac{x\lambda^\alpha}{(\lambda+x)^\alpha} \right] + \int_D^\infty \frac{\lambda^\alpha}{(\lambda+x)^\alpha} dx - D \left[-\frac{\lambda^\alpha}{(\lambda+x)^\alpha} \right] \\ &= 0 + \frac{D\lambda^\alpha}{(\lambda+D)^\alpha} + \left[-\frac{\lambda^\alpha}{(\alpha-1)(\lambda+x)^{\alpha-1}} \right] + 0 - \frac{D\lambda^\alpha}{(\lambda+D)^\alpha} \\ &= \frac{\lambda^\alpha}{(\alpha-1)(\lambda+D)^{\alpha-1}} \\ &= \frac{300^\alpha}{2(300 + 23.165)^2} \\ &= 129.27 \end{aligned}$$

$$\text{Hence } E[X - D | X > D] = 129.27/0.8 = 161.58$$

[5]**[Total Marks-9]****Solution 5 :**

Let the prior distribution of β be Gamma distribution with parameters α and λ .

$$E(\beta) = \frac{\alpha}{\lambda} = 200 \quad V(\beta) = \frac{\alpha}{\lambda^2} = 50^2$$

$$\text{Dividing Mean by Variance; } \lambda = \frac{200}{50^2} = 0.08,$$

$$\text{Thus } \alpha = 200 * \lambda = 200 * 0.08 = 16$$

The posterior distribution of β is then given by:

$$f(\beta|x) \propto f(x|\beta)f(\beta)$$

[2]

$$\propto e^{-\beta} * \beta^{240} * \beta^{\alpha-1} e^{-\lambda\beta}$$

$$\propto e^{-\beta} * \beta^{240} * \beta^{15} e^{-0.08\beta}$$

$$\propto \beta^{255} * e^{-1.08\beta}$$

this is similar to the pdf of Gamma distribution with parameters $\alpha' = 256$, $\lambda' = 1.08$.

Under quadratic loss, the Bayesian estimate is given by the mean of the posterior distribution, which is

$$\frac{\alpha'}{\lambda'} = \frac{256}{1.08} = 237.03$$

[Total Marks-7]

Solution 6 :

The probability that an $Exp(\lambda)$ distribution exceeds 100000 is just $e^{-1,00000\lambda}$. So the likelihood function is:

$$L(\lambda) = \lambda \exp(-\lambda x_1) \cdot \lambda \exp(-\lambda x_2) \cdot \dots \cdot \lambda \exp(-\lambda x_{226}) (\exp(-100000 \lambda))^{24}$$

$$= (\exp(-100000 \lambda))^{24} \exp(-\lambda(x_1 + x_2 + \dots + x_{226})) \lambda^{226}$$

Taking logs:

$$\log L = 226 \log \lambda - (100000 * 24 + (x_1 + x_2 + \dots + x_{226})) \lambda$$

Partial differentiating with respect to λ :

$$\text{First order partial derivative} = 226/\lambda - (100000*24 + (x_1 + x_2 + \dots + x_{226})).$$

Equating to zero, and solving for λ ,

$$\lambda = 226 / (100000*24 + (x_1 + x_2 + \dots + x_{226})) = 226/(100000*24 + 540 * 226) = 0.00008961.$$

Also, taking second order partial derivative of Log L with respect to λ , it gives:

$$- 226 / \lambda^2 < 0.$$

Hence this gives a maximum estimate.

[Total Marks-6]

Solution 7 :**(i) Advantages of using pseudo-random numbers**

We can generate the same sequence of pseudo-random numbers more than once (whereas for truly random numbers the values would have to be recorded in a lengthy table).

We only require a single routine to generate pseudo-random numbers (whereas for truly random numbers we need a lengthy table or a special piece of hardware).

[2]

(ii) Number of simulations to carry out and Measure of discrepancy

Monte Carlo simulation is usually undertaken to determine the expected value $\theta = E[X]$ of a random variable X connected with a particular stochastic model.

Simulations are performed until we have accumulated a total of n outputs $x_1, x_2, \dots, x_n$. The arithmetic average of these outputs ($\bar{\theta}$) is then used as an estimator of θ .

In a typical numerical simulation a tolerance level ε (the maximal acceptable value of error) is specified in advance.

Due to the stochastic nature of the Monte Carlo simulations, a confidence level $1 - \alpha$ (probability that the actual error is less than the tolerance level) should be also specified before the beginning of simulations.

Number of simulations is then determined such that the discrepancy between $\bar{\theta}$ and θ is less than ε with probability (at least) $1 - \alpha$.

There are two common ways to measure the discrepancy:

Absolute error

If the absolute error is used as the measure of discrepancy, then the following argument can be used to find the necessary number of simulations (n):

$$n > \frac{z_{\alpha/2}^2 \tau^2}{\varepsilon^2}$$

Where:

- n is the number of simulations to be carried out
- τ^2 is the estimated value of the variance
- Let $z_{\alpha/2}$ is the value obtained such that $P(-z_{\alpha/2} \leq N(0,1) \leq z_{\alpha/2}) = 1 - \alpha$

Relative Error

If the relative error is used as the measure of discrepancy, then the following argument can be used to find the necessary number of simulations (n):

$$n > \frac{z_{\alpha/2}^2 \tau^2}{\varepsilon^2 \theta^2}$$

Where:

- θ is the estimated value of the mean

Rest all the other items are as defined for absolute error above

[4]

[Total Marks-6]

Solution 8 :

(i) The main linear models used for modeling stationary time series are:

- **Autoregressive process (AR)**

An autoregressive process of order p (the notation $AR(p)$ is commonly used) is a sequence of random variables $\{X_t\}$ defined consecutively by the rule:

$$X_t = \mu + \alpha_1 (X_{t-1} - \mu) + \alpha_2 (X_{t-2} - \mu) + \dots + \alpha_p (X_{t-p} - \mu) + e_t$$

Thus the autoregressive model attempts to explain the current value of X as a linear combination of past values with some additional externally generated random variation.

- **Moving average process (MA)**

A moving average process of order q , denoted $MA(q)$, is a sequence $\{X_t\}$ defined by the rule:

$$X_t = \mu + e_t + \beta_1 e_{t-1} + \dots + \beta_q e_{t-q}$$

The moving average model explains the relationship between the X_t as an indirect effect, arising from the fact that the current value of the process results from the recently past random error terms as well as the current one.

Autoregressive moving average process (ARMA)

The two basic processes (AR and MA) can be combined to give an autoregressive moving average, or ARMA, process. The defining equation of an $ARMA(p,q)$ process is:

$$X_t = \mu + \alpha_1 (X_{t-1} - \mu) + \alpha_2 (X_{t-2} - \mu) + \dots + \alpha_p (X_{t-p} - \mu) + e_t + \beta_1 e_{t-1} + \dots + \beta_q e_{t-q}$$

[6]

(ii)

a. This is $MA(1)$ process and hence it is stationary (as it is the sum of stationary white noise terms). Therefore we can classify it as $ARIMA(0,0,1)$.

b. This is an $ARMA(2,3)$ process.

This process cannot be differenced, so to be able to classify it as an $ARIMA(2,0,3)$, we must check that it is $I(0)$, i.e. stationary.

Since $(1-1.4B^2) X_t = \varepsilon_t + 0.5 \varepsilon_{t-3}$, the characteristic equation of the AR terms is:

$$\phi(\lambda) = 1 - 1.4 \lambda^2 = 0 \quad \Rightarrow \lambda = \pm 0.8452$$

Since both of the roots are less than one in magnitude the process is not stationary and so we cannot classify it as an $ARIMA(2,0,3)$.

It is a non-stationary $ARMA(2,3)$ process.

c. This is an ARMA(2,1) process.

This process can be differenced as follows:

$$X_t - 1.4X_{t-1} + 0.4X_{t-2} = \varepsilon_t + \varepsilon_{t-1}$$

$$(X_t - X_{t-1}) - 0.4(X_{t-2} - X_{t-3}) = \varepsilon_t + \varepsilon_{t-1}$$

$$\Delta X_t - 0.4 \Delta X_{t-1} = \varepsilon_t + \varepsilon_{t-1}$$

This process cannot be differenced again, so to be able to classify it as an ARIMA(1,1,1), we must check whether this differenced process is stationary (i.e. the original process is I (1)).

Since $(1 - 0.4B) \Delta X_t = \varepsilon_t + \varepsilon_{t-1}$, the characteristic equation of the differenced AR terms is:

$$\phi(\lambda) = 1 - 0.4\lambda = 0 \Rightarrow \lambda = 2.5$$

Since the root is greater than one in magnitude the differenced process is stationary (i.e. the original process is I (1)). Therefore, we can classify it as an ARIMA(1,1,1) .

[4]

(iii) The process is stationary as it the sum of stationary white noise terms, so we can calculate the autocovariance function (ignoring the 3.1's as they will not affect the results and noting that $\gamma_k = \gamma_{-k}$):

$$\begin{aligned} \gamma_0 &= \text{Cov}(X_t, X_t) = \text{var}(X_t) \\ &= \text{Cov}(\varepsilon_t + 0.25\varepsilon_{t-1} + 0.5\varepsilon_{t-2} + 0.25\varepsilon_{t-3}, \varepsilon_t + 0.25\varepsilon_{t-1} + 0.5\varepsilon_{t-2} + 0.25\varepsilon_{t-3}) \\ &= \sigma^2 + 0.25^2\sigma^2 + 0.5^2\sigma^2 + 0.25^2\sigma^2 \\ &= 1.375\sigma^2 \end{aligned}$$

$$\begin{aligned} \gamma_{+/-1} &= \text{Cov}(X_t, X_{t-1}) \\ &= \text{Cov}(\varepsilon_t + 0.25\varepsilon_{t-1} + 0.5\varepsilon_{t-2} + 0.25\varepsilon_{t-3}, \varepsilon_{t-1} + 0.25\varepsilon_{t-2} + 0.5\varepsilon_{t-3} + 0.25\varepsilon_{t-4}) \\ &= 0.25\sigma^2 + (0.5)(0.25)\sigma^2 + (0.5)(0.25)\sigma^2 \\ &= 0.5\sigma^2 \end{aligned}$$

$$\begin{aligned} \gamma_{+/-2} &= \text{Cov}(X_t, X_{t-2}) \\ &= \text{Cov}(\varepsilon_t + 0.25\varepsilon_{t-1} + 0.5\varepsilon_{t-2} + 0.25\varepsilon_{t-3}, \varepsilon_{t-2} + 0.25\varepsilon_{t-3} + 0.5\varepsilon_{t-4} + 0.25\varepsilon_{t-5}) \\ &= 0.5\sigma^2 + 0.25^2\sigma^2 \\ &= 0.5625\sigma^2 \end{aligned}$$

$$\begin{aligned} \gamma_{+/-3} &= \text{Cov}(X_t, X_{t-3}) \\ &= \text{Cov}(\varepsilon_t + 0.25\varepsilon_{t-1} + 0.5\varepsilon_{t-2} + 0.25\varepsilon_{t-3}, \varepsilon_{t-3} + 0.25\varepsilon_{t-4} + 0.5\varepsilon_{t-5} + 0.25\varepsilon_{t-6}) \\ &= 0.25\sigma^2 \end{aligned}$$

$$\gamma_{+/-k} = 0 \text{ for } |k| > 3$$

Since $\rho_k = \gamma_k / \gamma_0$, the autocorrelation function is:

$$\rho_0 = 1$$

$$\rho_{+/-1} = 0.364$$

$$\rho_{+/-2} = 0.409$$

$$\rho_{+/-3} = 0.182$$

$$\rho_{+/-k} = 0 \text{ for } |k| > 3$$

[5]

[Total Marks-15]

Solution 9 :

(i) The Bornhuetter-Ferguson method combines the estimated loss ratio with a projection method.

Steps involved in calculating using the Bornhuetter-Ferguson method are as follows:

- Determine the initial estimate of the total ultimate claims from each origin year using premiums and loss ratios.
- Divide these estimates by projection factors (f) determined, in a normal manner, from a claims development table. These are effectively estimates of the claims that should have developed to date.
- Subtract these amounts from the corresponding total ultimate claims figures to give an estimate of the amount of claims that are yet to develop.

The advantage of this method is that it improves on the crude use of a loss ratio by taking account of the information provided by the latest development pattern of the claims, whilst the addition of the loss ratio to a projection method serves to add some stability against distortions in the development pattern.

[4]

(ii) We need the cumulative claims data:

		Development Year				
		1	2	3	4	5
Accident Year	1	4,900	7,350	9,077	9,641	9,741
	2	5,705	9,212	10,992	11,729	
	3	6,745	10,207	11,707		
	4	6,514	10,214			
	5	8,097				

The ultimate loss ratio is $9,741 / 11,100 = 0.87757$

Next we calculate the expected end of year figures (the initial ultimate liability):

$$2 : 0.87757 * 13,677 = 12,002$$

$$3 : 0.87757 * 15,585 = 13,677$$

$$4 : 0.87757 * 16,381 = 14,375$$

$$5 : 0.87757 * 16,507 = 14,486$$

The development factors are:

Year 4 to Year 5

$$9741 / 9641 = 1.01037$$

Year 3 to Year 4

$$(9641 + 11,729) / (9,077 + 10,992) = 1.06483$$

Year 2 to Year 3

$$(9,077 + 10,992 + 11,707) / (7,350 + 9,212 + 10,207) = 1.18704$$

Year 1 to Year 2

$$(7,350 + 9,212 + 10,207 + 10,214) / (4,900 + 5,705 + 6,745 + 6,514) = 1.54974$$

The emerging liabilities for each year are:

2: 12,002 * (1 - 1 / 1.01037)	= 123
3: 13,677 * (1 - 1 / 1.01037 * 1.06483)	= 965
4: 14,375 * (1 - 1 / 1.01037 * 1.06483 * 1.18704)	= 3,119
5: 14,486 * (1 - 1 / 1.01037 * 1.06483 * 1.18704 * 1.54974)	= 7,167

Given that these are claims paid (rather than incurred), we don't need to calculate the revised ultimate liability to get the reserve we can just total up the emerging liabilities:

$$123 + 965 + 3119 + 7167 = 11,374$$

The assumptions underlying the above calculation are:

- Payments from each origin year will develop in the same way.
- Weighted average past inflation will be repeated in the future.
- The first year is fully run-off.
- The estimated loss ratio is appropriate.

[6]

[Total Marks-10]
